Production System in AI
Intelligent Agent
Problem Solving in AI

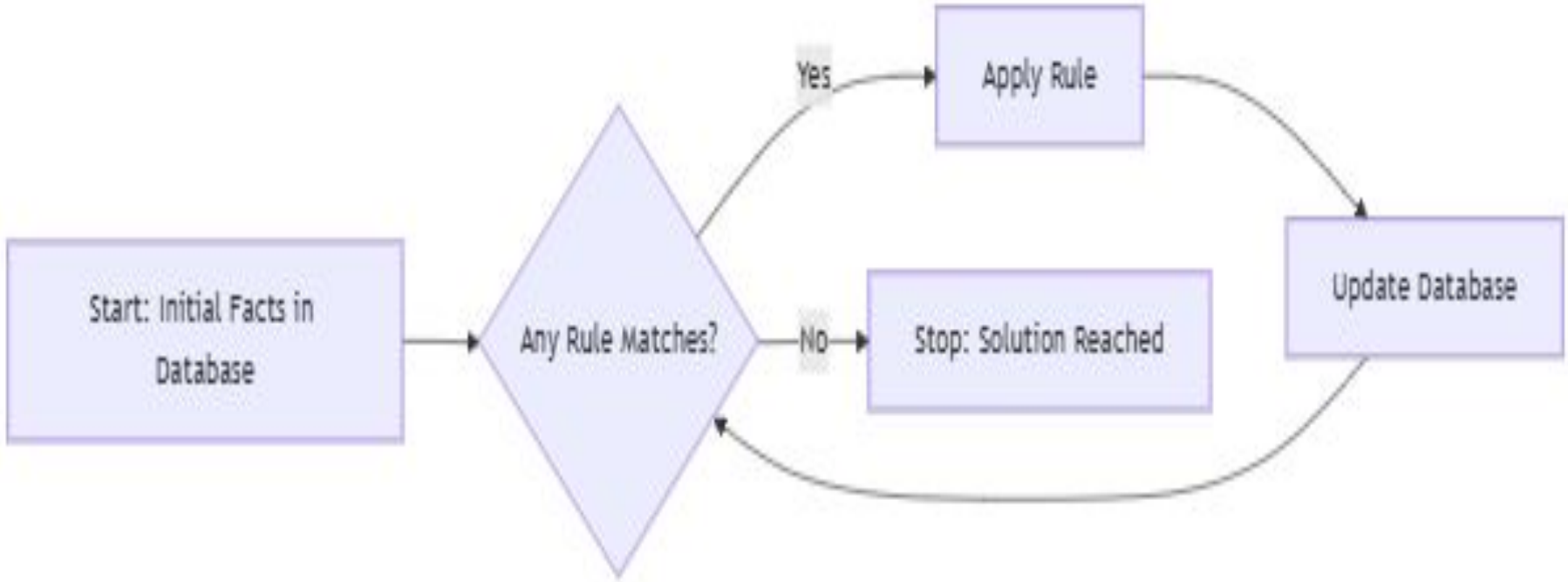
Production Systems

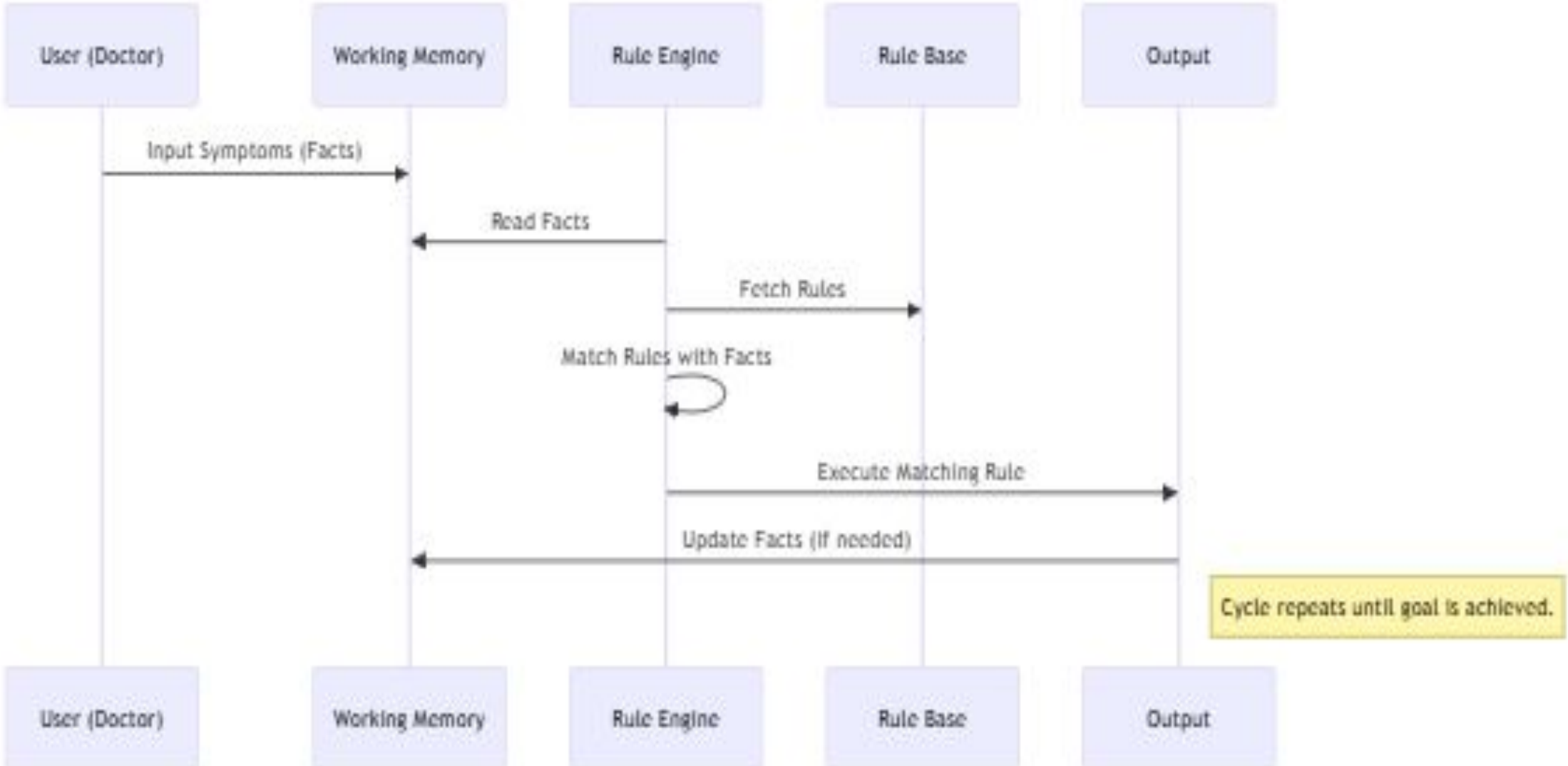
A production system in AI is a framework for building knowledge-based problem-solving models. It consists of:

1) **A set of rules (productions):** Each rule has a condition (if part) and an action or outcome (then part).
2) **A database (working memory):** Current state information used to match rule conditions.
3) **A control strategy:** Decides which rule to apply when more than one can be triggered.
4) **Rule Applier**

Example:
Rule: If it is raining, then take an umbrella.

Real Life Example
Medical Diagnosis: If symptoms include high fever and rash, then suspect measles.
E-commerce: If a customer has purchased a laptop, recommend laptop bags.




User (Doctor)
Working Memory
Rule Engine
Rule Base
Output
Input Symptoms (Facts)
Read Facts
Fetch Rules
Match Rules with Facts
Execute Matching Rule
Update Facts (If needed)
Cycle repeats until goal is achieved.
User (Doctor)
Working Memory
Rule Engine
Rule Base
Output

Types of Production Systems

Type	Description	Example
Monotonic	Rules do not delete facts; knowledge only grows.	Mathematical theorem proving.
Non-monotonic	Rules can both add and remove facts.	Dynamic expert systems.
Deterministic	Same input always leads to same output/rule.	Straightforward calculations.
Non-deterministic	May yield different results for same inputs.	Games with random events.

Problem Characteristics

Def: How a problem is structured for an AI system, affecting which techniques are suitable.

Problems in AI can be characterized by:

- Decomposability: Can the problem be broken into smaller, independent subproblems?
- Solution Steps: Is a sequence of actions required?
- Predictability: Are the outcomes of actions predictable or uncertain?
- Observability: Is the agent's information about the state complete or partial?
- Static vs Dynamic: Does the problem change while the agent is solving it?

Example: Chess is decomposable and deterministic with sequential solution steps.

Real Life Example:
Route Planning: Large, decomposable, dynamic, sequential.
Sudoku: Fully observable, deterministic, static, discrete.

Characteristic	Route Planning	Sudoku
Decomposable	Yes	Yes
Observability	Fully	Fully
Deterministic	Semi (traffic varies)	Yes
Static/Dynamic	Dynamic	Static
Discrete/Continuous	Continuous	Discrete

Intelligent Agents

- Artificial intelligence is defined as a **study of rational agents**.
- A rational agent could be anything which **makes decisions**, as a person, firm, machine, or software.
- It carries out an action with the best outcome after considering past and current percepts(agent's perceptual inputs at a given instance).

Intelligent Agents

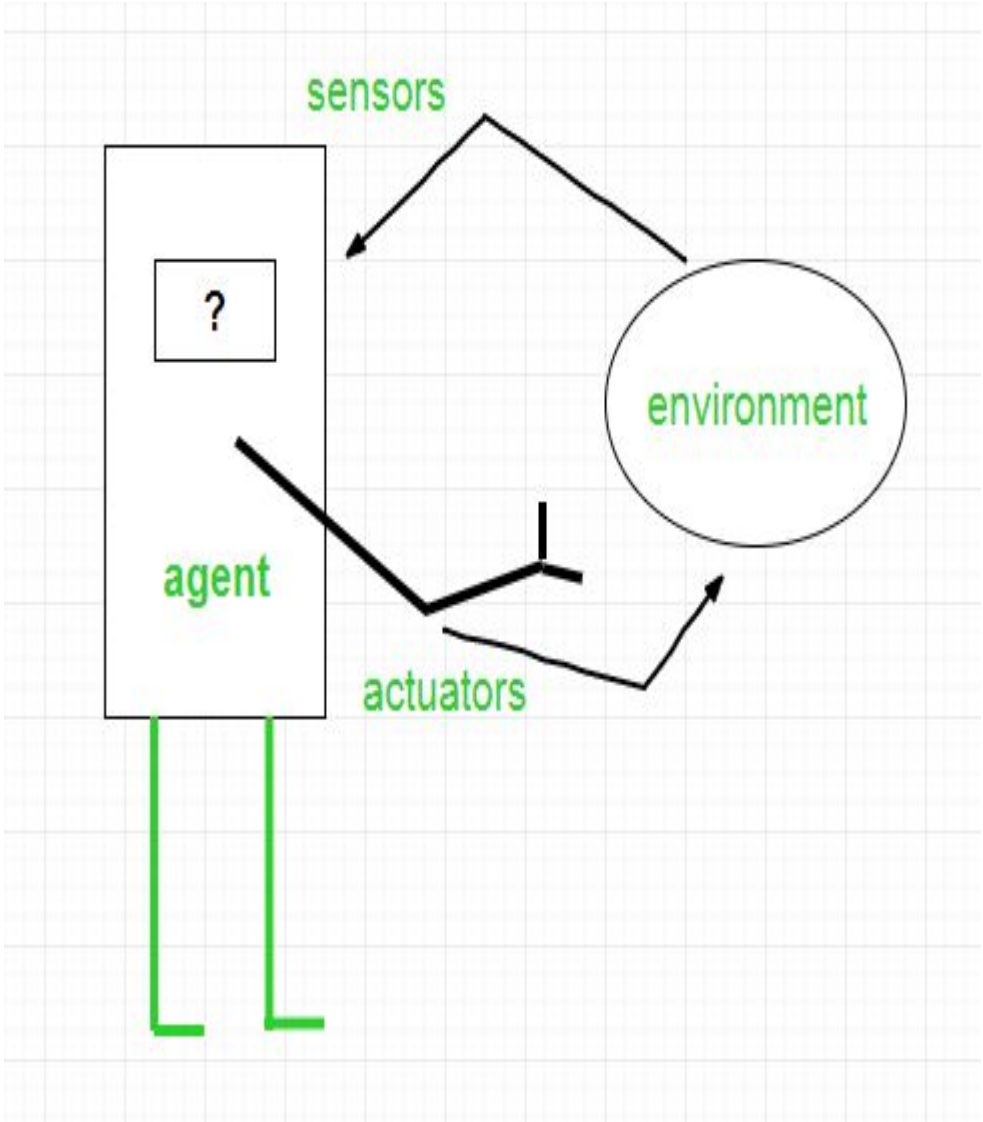
- An AI system is composed of an
  - **Agent and its**
  - **Environment**.
- The agents act in their environment.
- The environment may contain other agents.
- An agent is anything that can be viewed as :
  - perceiving its environment through **sensors** and
  - acting upon that environment through **actuators**
- **Note** : Every agent can perceive its own actions (but not always the effects)

Intelligent Agents- Basics Terminology

- To understand the structure of Intelligent Agents, we should be familiar with
- ***Architecture*** **and**
- ***Agent Program*****.**
- **Architecture** is the machinery that the agent executes on.

  It is a **device with sensors and actuators**,

  for example : a robotic car, a camera, a PC.
- **Agent program** is an implementation of an agent function.
- An **agent function** is a map from the percept sequence(history of all that an agent has perceived till date) to an action.
- **Agent = Architecture + Agent Program**



Intelligent Agents- Examples

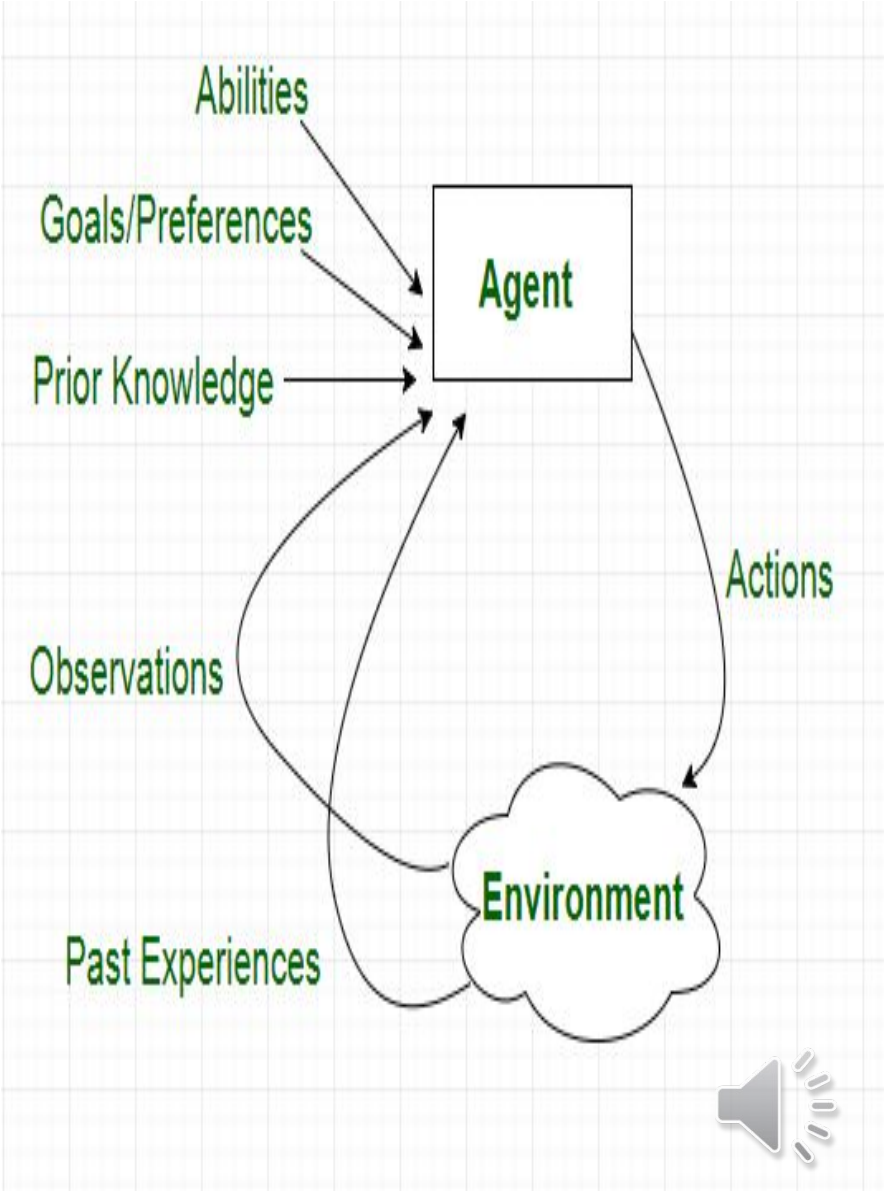
Examples of Agent:-

A **Software agent** has Keystrokes, file contents, received network packages which act as sensors
and displays on the screen, files, sent network packets acting as actuators

.

A **Human agent** has eyes, ears, and other organs which act as sensors and hands, legs, mouth, and other body parts acting as actuators.

A **Robotic agent** has Cameras and infrared range finders which act as sensors and various motors acting as actuators.



Intelligent Agents- Types of Agent

Types of Agents –

Agents can be grouped into four classes based on their degree of perceived intelligence and capability :

- Simple Reflex Agents and Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agent

Types of Intelligent Agent- Simple Reflex Agents

Ignore the rest of the percept history and **act only on** the basis of the **current percept**.

- Percept history is the history of all that an agent has perceived till date. The agent function is based on the **condition-action rule**.
- **A condition-action rule** is a rule that maps a state i.e, condition to an action. If the condition is true, then the action is taken, else not. This agent function only succeeds when the environment is fully observable.
- For simple reflex agents operating in partially observable environments, infinite loops are often unavoidable. It may be possible to escape from infinite loops if the agent can randomize its actions.

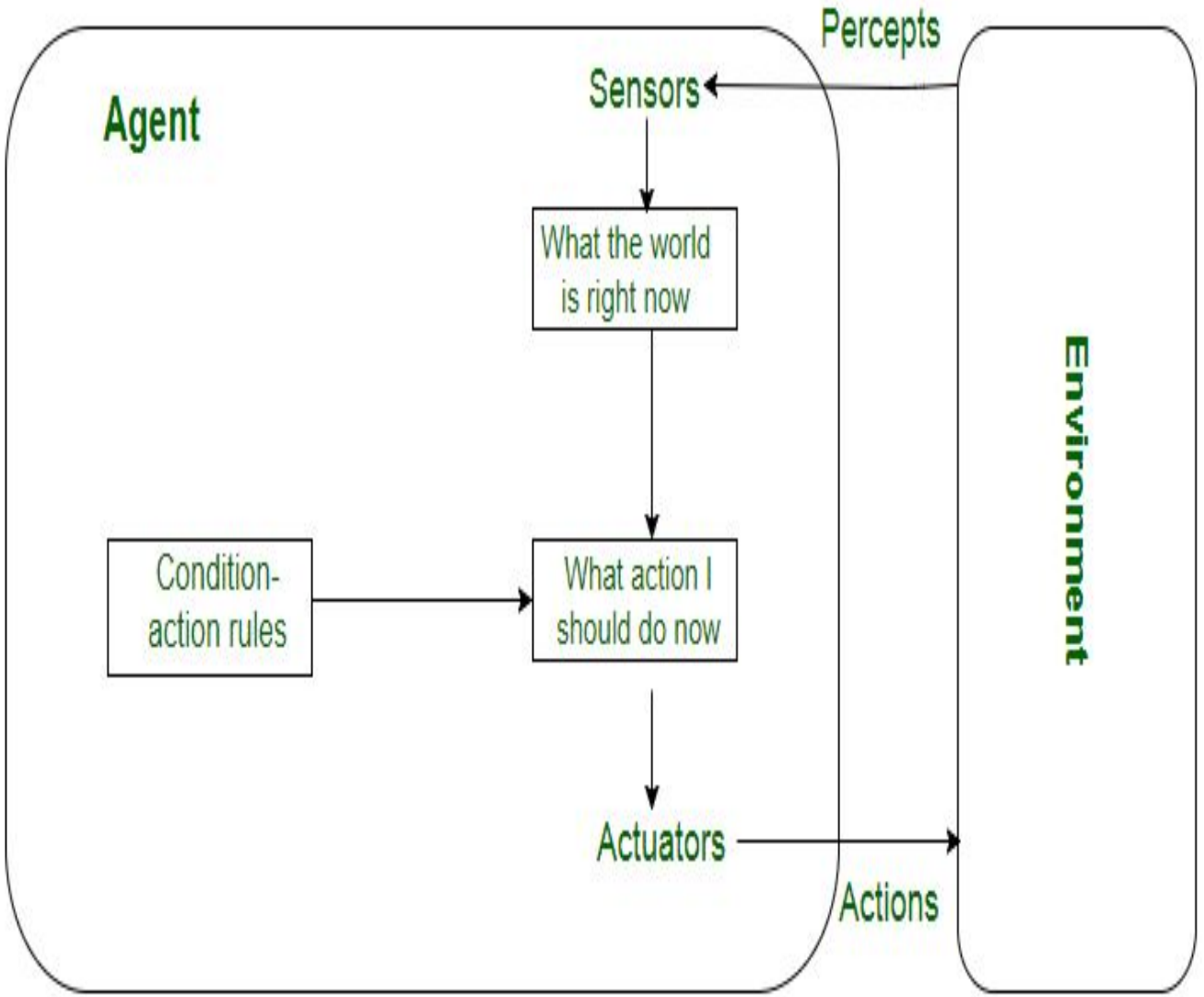


Types of Intelligent Agent- Simple Reflex Agents

Problems with Simple reflex agents are :

- Very limited intelligence.
- No knowledge of non-perceptual parts of state.
- Usually too big to generate and store.

If there occurs any change in the environment, then the collection of rules need to be updated.



Types of Intelligent Agent- Model-based reflex agents

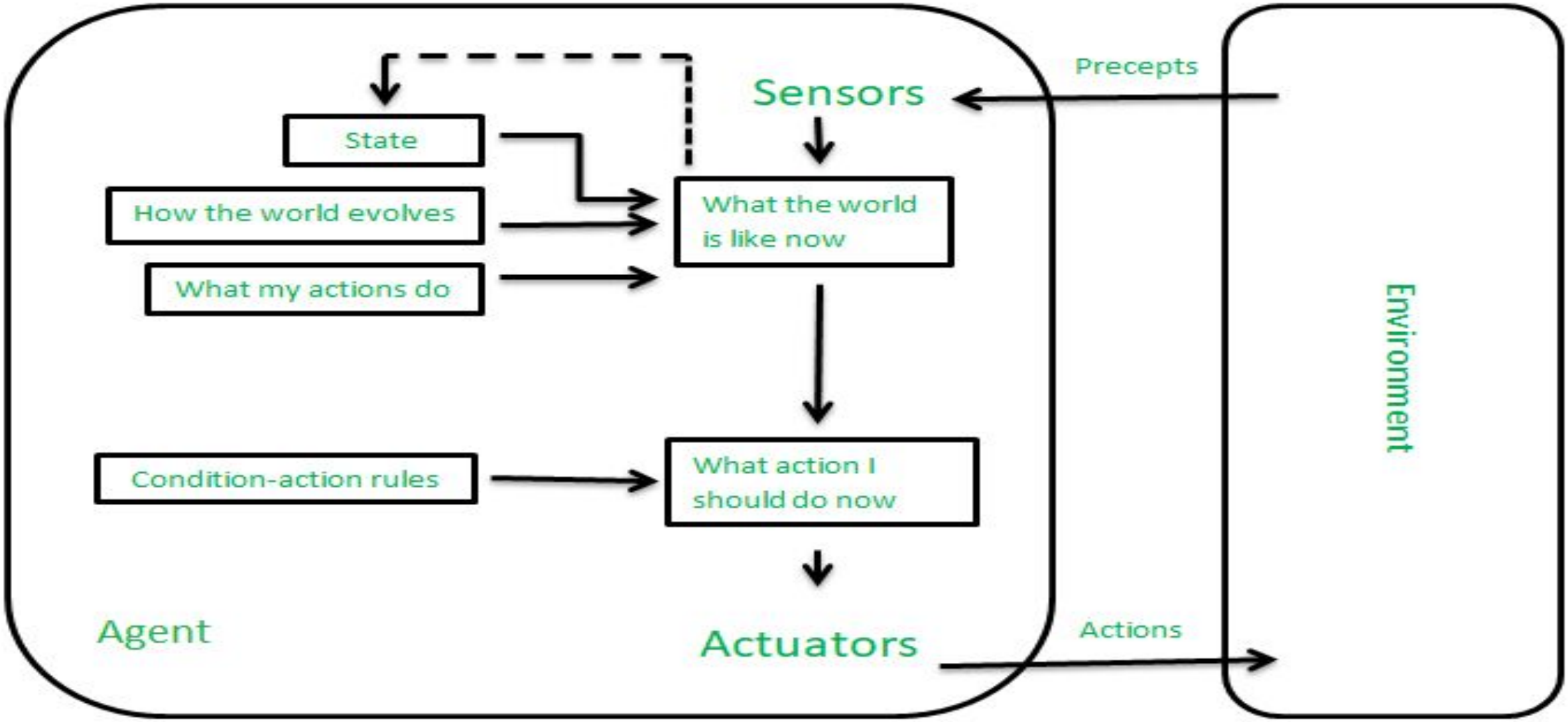
It works by finding a rule whose condition matches the current situation.
A model-based agent can handle **partially observable environments** by use of model about the world. The agent has to keep track of **internal state** which is adjusted by each percept and that depends on the percept history.

The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen. Updating the state requires information about :

- how the world evolves in-dependently from the agent, and
- how the agent actions affects the world.



Types of Intelligent Agent- Model-based reflex agents

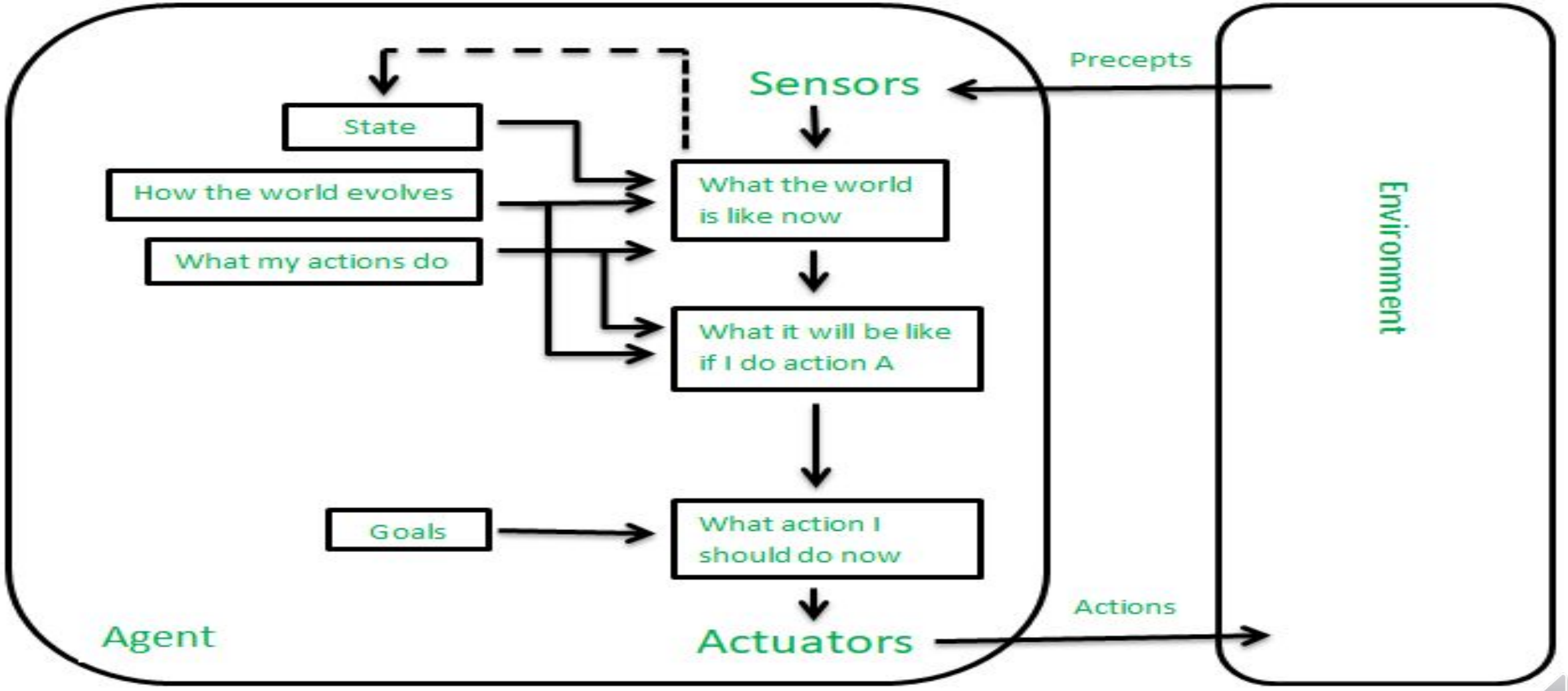


Types of Intelligent Agent- Goal-based agents

- These kind of agents take decision based on how far they are currently from their **goal** (description of desirable situations).
- Their every action is intended to reduce its distance from the goal.
- This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state.
- The knowledge that supports its decisions is represented explicitly and can be modified, which makes these agents more flexible. They usually require search and planning.
- The goal-based agent's behavior can easily be changed



Types of Intelligent Agent- Goal-based agents

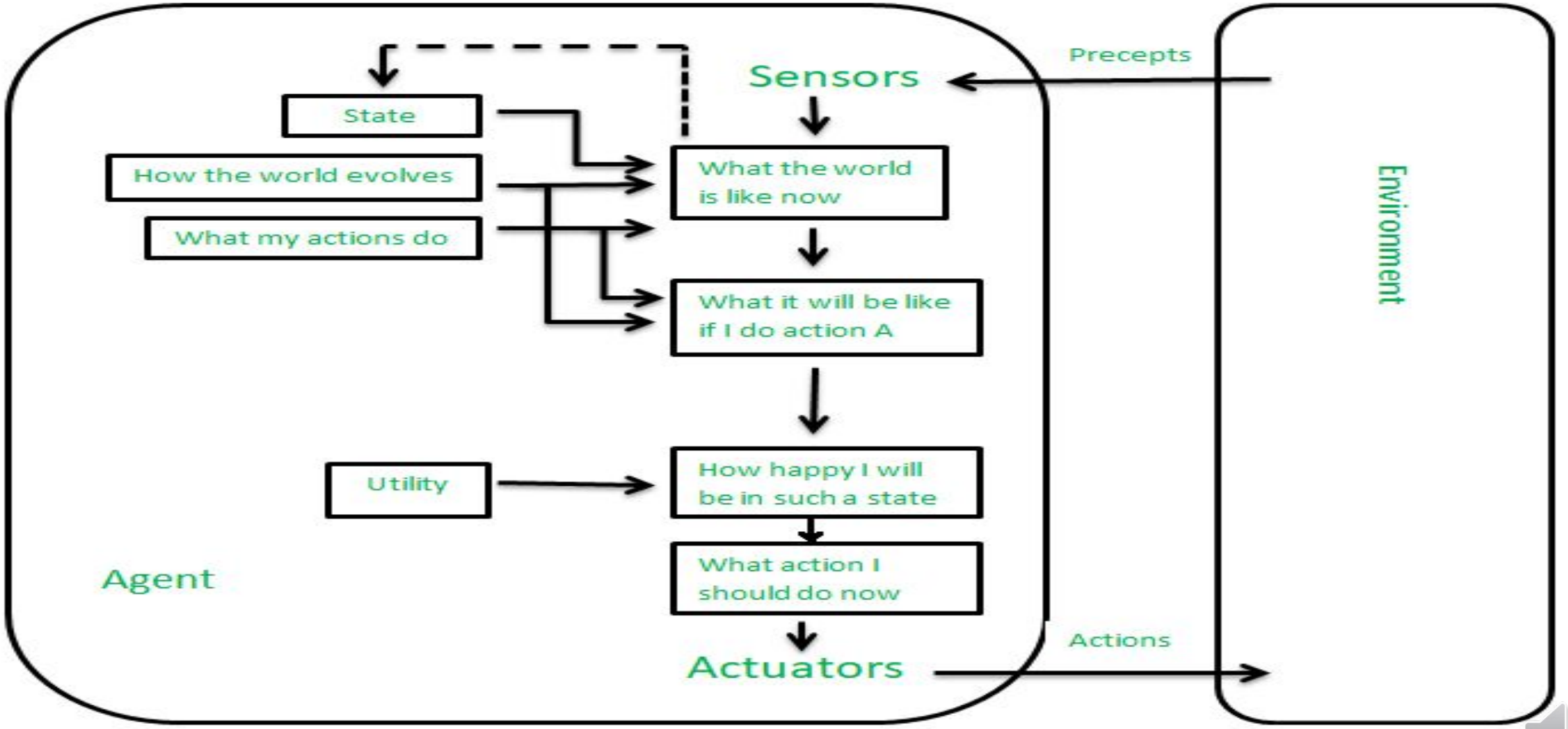


Types of Intelligent Agent- Utility-based agents

- The agents which are developed having their end uses as building blocks are called **utility-based agents**.
- When there are multiple possible alternatives, then to decide which one is best, utility-based agents are used.
- They choose actions based on a **preference (utility)** for each state. Sometimes achieving the desired goal is not enough. We may look for a quicker, safer, cheaper trip to reach a destination. Agent happiness should be taken into consideration. Utility describes how **"happy"** the agent is. Because of the uncertainty in the world, a utility agent chooses the action that maximizes the expected utility. A utility function maps a state onto a real number which describes the associated degree of happiness.



Types of Intelligent Agent- Utility-based agents



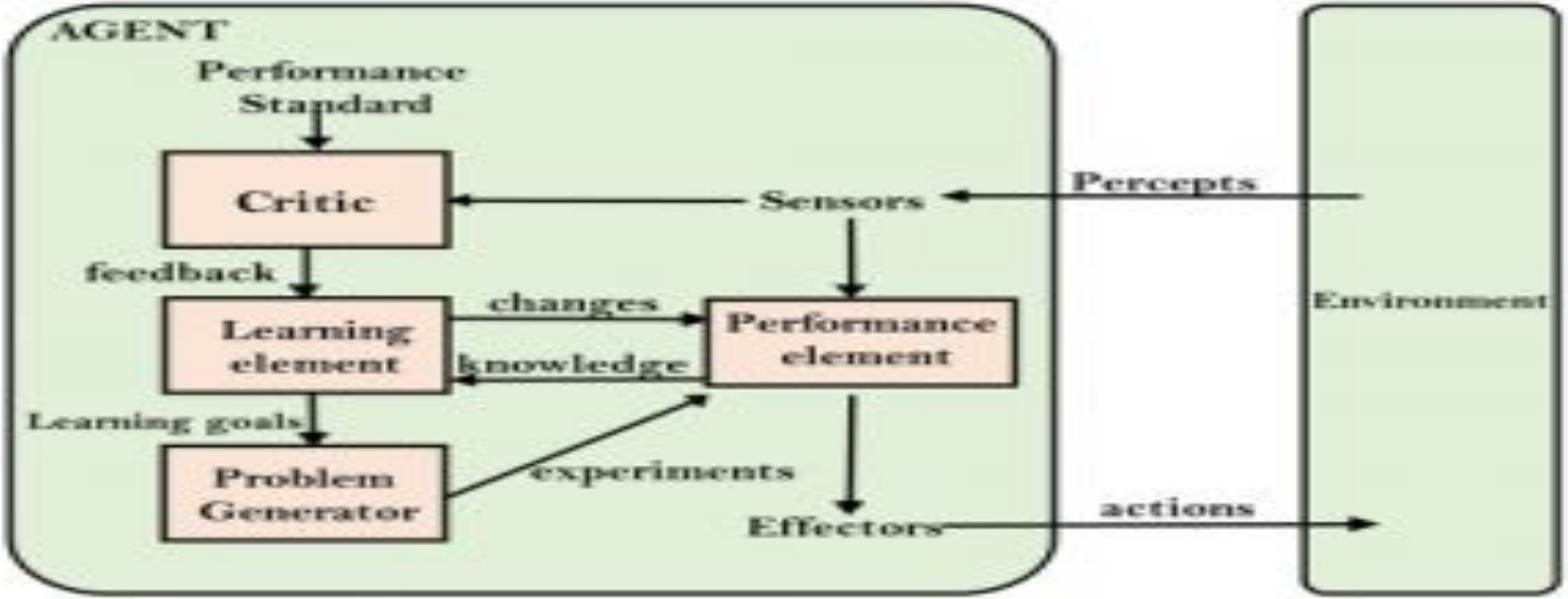
Types of Intelligent Agent- Learning Agent

A learning agent in AI is the type of agent which can learn from its past experiences or it has learning capabilities.
It starts to act with basic knowledge and then able to act and adapt automatically through learning.

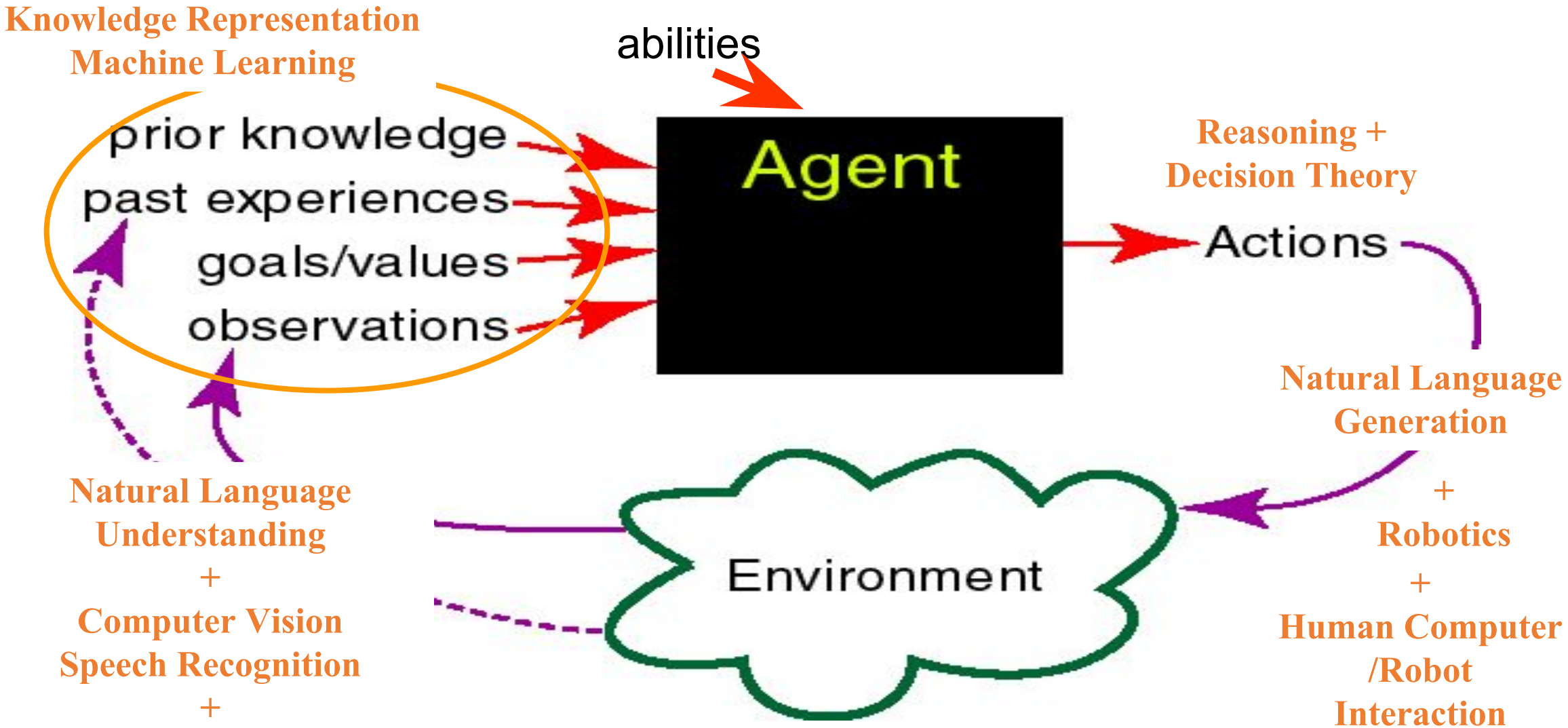
A learning agent has mainly four conceptual components, which are:

- **Learning element:** It is responsible for making improvements by learning from the environment
- **Critic:** Learning element takes feedback from critic which describes how well the agent is doing with respect to a fixed performance standard.
- **Performance element:** It is responsible for selecting external action
- **Problem Generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Types of Intelligent Agent- Learning Agent



Intelligent Agents in the World

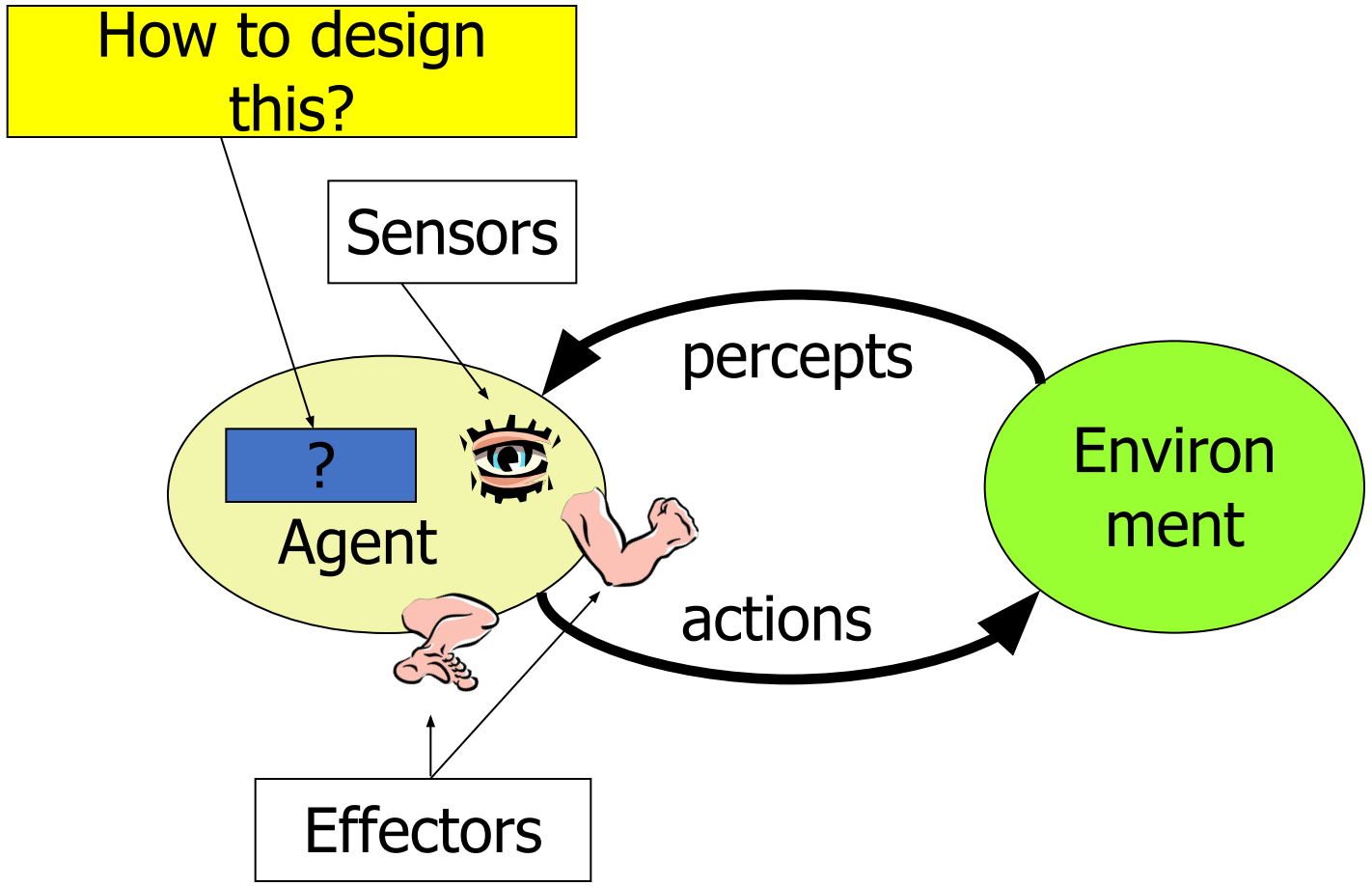


Why do we need intelligent agents?

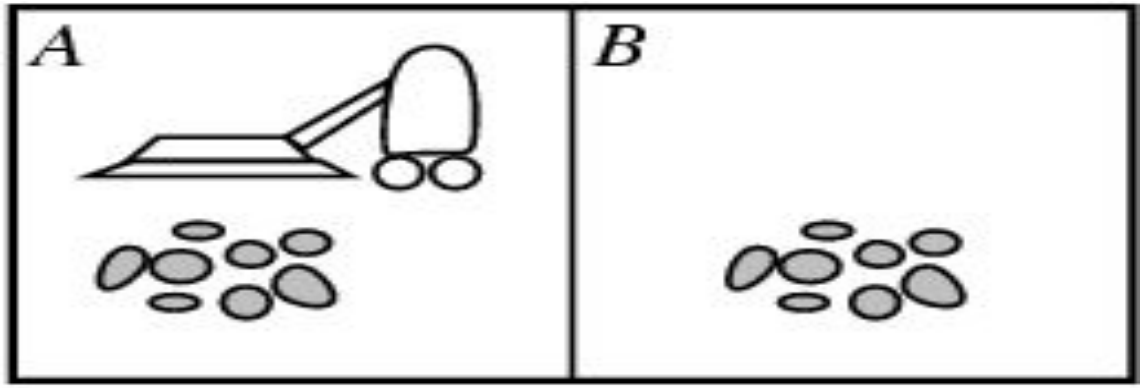
- Go where humans can't go (**dangerous/impossible for humans**)
- Do unpleasant work (**tedious/boring**)
- Higher **efficiency**
- Complex problems that have to be **solved quickly**
- **Entertainment**
- More **accurate simulation and predictions** of human behaviour
- Perform a task **autonomously**



Typical Intelligent Agent- Rational Agents



Vacuum-cleaner world



- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*
- *Agent's function → look-up table*
  - *For many agents this is a very large table*

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
⋮	⋮

PEAS

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- Consider,

  e.g., the task of designing an automated taxi driver:

  - Performance measure
  - Environment
  - Actuators
  - Sensors



PEAS

Must first specify the setting for intelligent agent design Consider,

e.g., the task of **designing an automated taxi driver:**

- **Performance measure:** Safe, fast, legal, comfortable trip, maximize profits
- **Environment:** Roads, other traffic, pedestrians, customers
- **Actuators:** Steering wheel, accelerator, break, signal, horn
- **Sensors:** Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard



PEAS

Describe the PEAS following agents

Agent: Medical diagnosis system

Agent: Part-picking robot

Agent: Interactive English tutor



PEAS

Agent: Medical diagnosis system

- **Performance measure:** Healthy patient, minimize costs, lawsuits
- **Environment:** Patient, hospital, staff
- **Actuators** : Screen display (questions, tests, diagnoses, treatments, referrals)
- **Sensors** : Keyboard (entry of symptoms, findings, patient's answers)



PEAS

Agent: Part-picking robot

- **Performance measure:** Percentage of parts in correct bins
- **Environment**: Conveyor belt with parts, bins
- **Actuators:** Jointed arm and hand
- **Sensors:** Camera, joint angle sensors



PEAS

Agent: Interactive English tutor

- **Performance measure:** Maximize student's score on test
- **Environment:** Set of students
- **Actuators:** Screen display (exercises, suggestions, corrections)
- **Sensors:** Keyboard



Environment

There are various ways to define environments in which the agent operates.

- Fully observable (accessible) vs Partially observable (inaccessible)
- Deterministic vs stochastic
- Episodic Vs Sequential
- Static Vs Dynamic
- Discrete Vs Continuous
- Single agent Vs Multi-agent



Fully observable (accessible) vs Partially observable (inaccessible)

In **fully observable** type agent sensor detects all aspect of environment relevant to choice of action

Example of fully observable environment :-

Chess: the board is fully observable as are opponents moves, Image analyzer, Crossword puzzle

Partially observable environment could be partially observable due to noisy, inaccurate or missing sensors or inability to measure everything which is needed

Example of partially observable environment

Driving: what is around the next bend is not observable

Deterministic vs. non-deterministic (stochastic)

- A **deterministic** environment is one in which any action has a single guaranteed effect — there is no uncertainty about the state that will result from performing an action
  - Crossword puzzle
  - Internet shopping
- The physical world can to all intents and purposes be regarded as non-deterministic
- **Non-deterministic** environments present greater problems for the agent designer
  - Taxi driver

Static vs. dynamic

- A **static** environment is one that can be assumed to remain unchanged except by the performance of actions by the agent
  - Crossword puzzle
- A **dynamic** environment is one that has other processes operating on it, and which hence changes in ways beyond the agent's control
  - Part picking robot
- Other processes can interfere with the agent's actions (as in concurrent systems theory)
- The physical world is a highly dynamic environment

Discrete vs. continuous

- An environment is **discrete** if there are a fixed, finite number of actions and percepts in it
  - Chess game
- **Continuous** environments have a certain level of mismatch with computer systems
  - Taxi driving

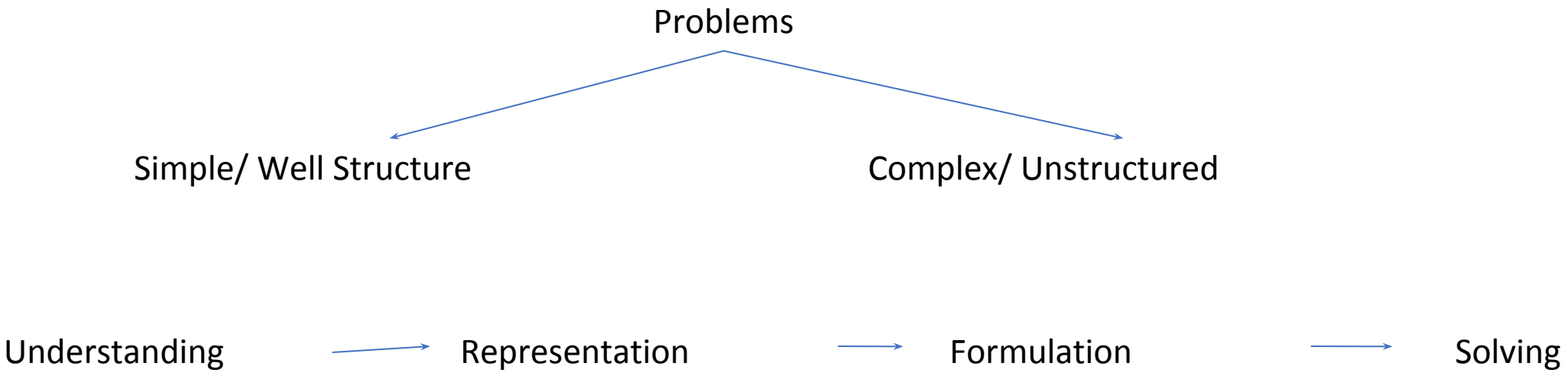
Single agent vs. Multiagent

- An agent operating by itself in an environment in single agent
  - Part picking robot

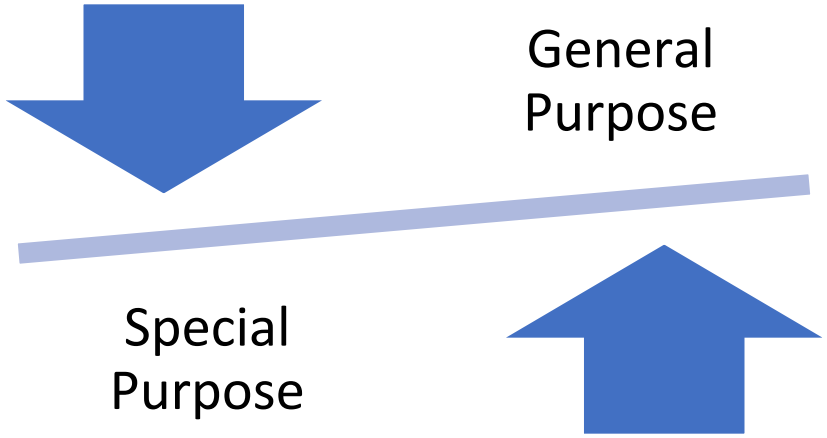
- **Multi agent** is when other agents are present other agent is anything that changes from step to step or other agent is something which must sense and act
  - Game playing like Chess

Problem Solving

- Finding out answer for some unknown situations.



Methods for Problem Solving



Means-end analysis

Present situation is compared with goal to detect difference

Eg. BFS, DFS searching algorithm and its application

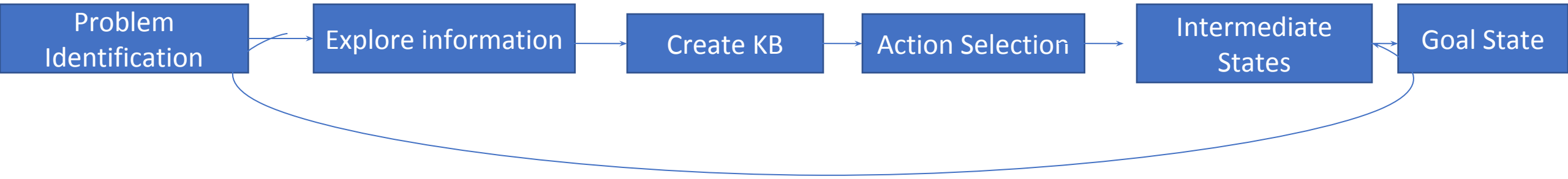
Particular problem modeled with various assumption, specific to that problem

Eg. Searching algorithm uses heuristic function

Behaves like expert system, assists in planning and decision making

Problem Solving Process

- what is problem: journey from unknown to known.
- Journey is nothing but process.
  - May vary from individual to individual



Problem solving process

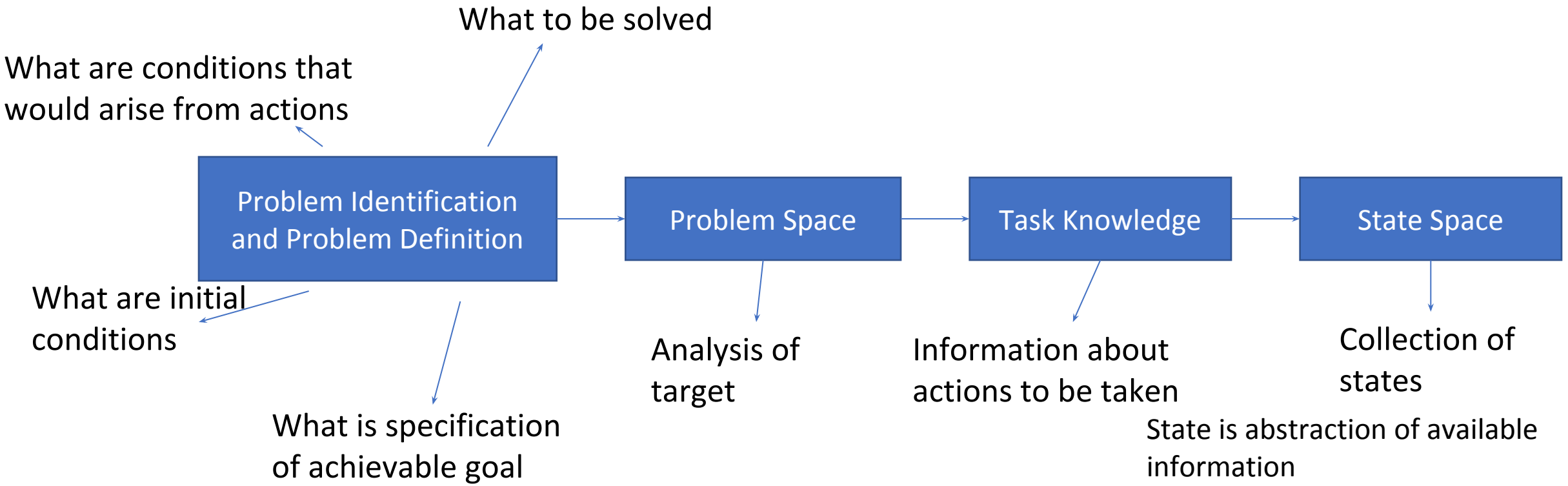
Eg. Route finding

Problem Solving Process

- General Problem-solving techniques involve:

1. Problem Identification
2. Problem analysis and representation
3. Planning
4. Execution
5. Evaluation of solution
6. Consolidating gains

Problem Formulation



Eg. 4 puzzle problem

Problem Analysis and Representation

- Performance of solution depends on Problem Representation
- PROBLEM DEFINITION should satisfy:
  - Compactness: must be able to restrict and define boundaries clearly
  - Utility: must be compatible with good solution algorithm
  - Soundness: should not report false
  - Completeness: should not loose any information
  - Generality: should be able to capture maximum instance of problem
  - Transparency: reasoning with representation efficiently

Problem Representation

- Problem representation consists of domain description and solver algorithm.
- Entities should be there:
  1. Initial State
  2. Goal State
  3. Actions
  4. Path Cost

What is State Space Search?

Pre-requisite of UNIT-II

What is State Space?

About “State Space”

- A SET, consisting of all possible CONFIGURATIONS of a SYSTEM
- Can be a finite / infinite set
- Contains the START state and the GOAL state
- One State can change to its NEIGHBORING State with 1 VALID MOVE
  - NEIGHBOURING state “s1” of a state “s” is a state which is possible with 1 valid MOVE from state “s”
- Valid moves change a state to its neighboring state
  - A State can have multiple Neighboring States as per number of Valid Moves from that State

State Space

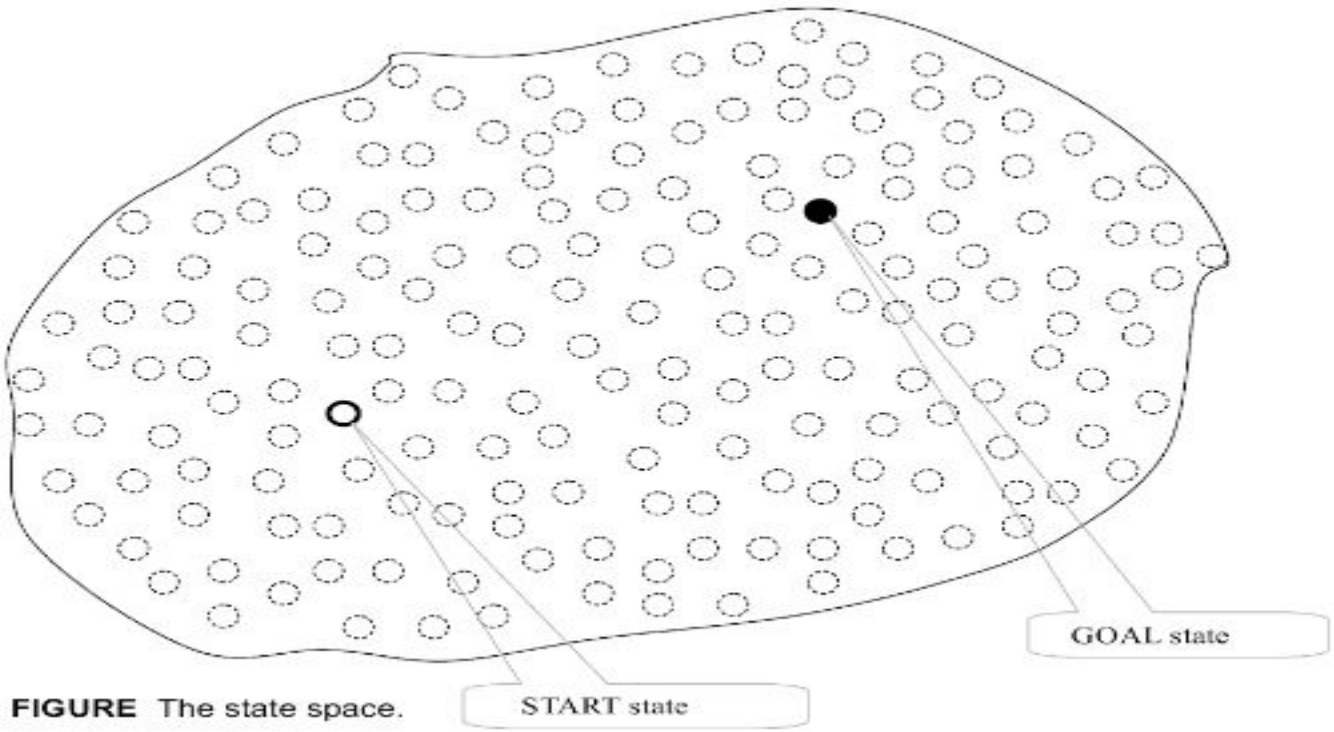
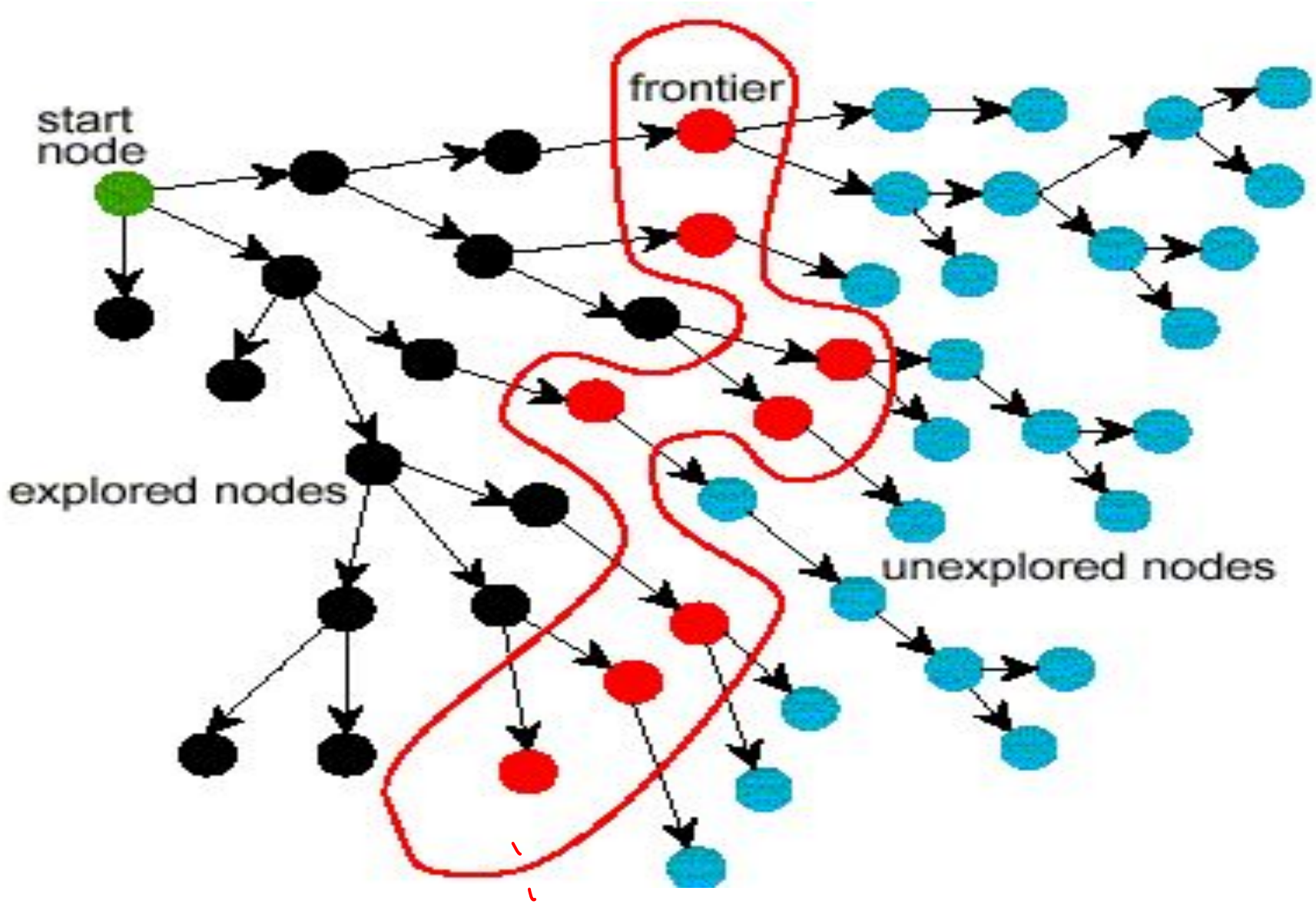
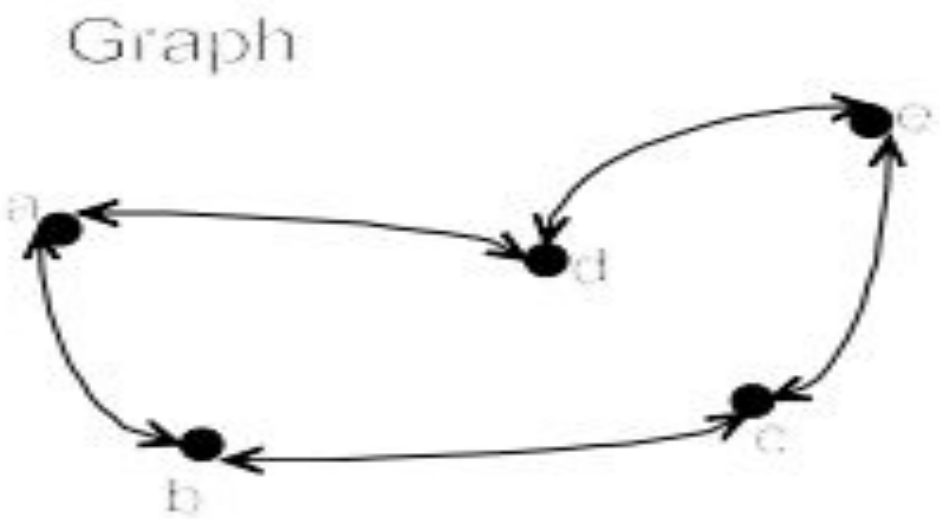


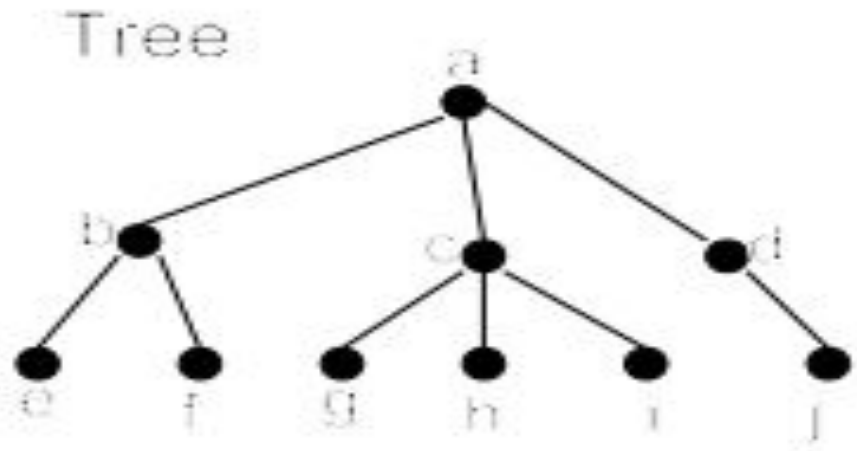
FIGURE The state space.

Problem Solving by Graph Searching



Graph vs Tree





Graph vs. Tree